

Web Programming II

Department of Data Science

Dr. UAP Ishanka

Learning Outcomes

At the completion of this course student will be able to,

- Describe concept of server side scripting using PHP programming
- Develop and design appropriate decision and looping structures, procedures and functions
- Create a program to read given data from an external server-side database
- Load persistent data onto a dynamic Web page
- Create programs to dynamically load elements as the user browses the site given dynamic parameters for a Web page

Evaluation

Quizzes (04) - 20%

Project (01) - 20%

End Exam - 60%

References

- Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5 (Learning PHP, MYSQL, Javascript, CSS & HTML5) , Robin Nixon, 5th Edition, 2020
- PHP and MySQL for Dynamic Web Sites: Visual QuickPro Guide 5th Edition, Larry Ullman, 2017

What is PHP?

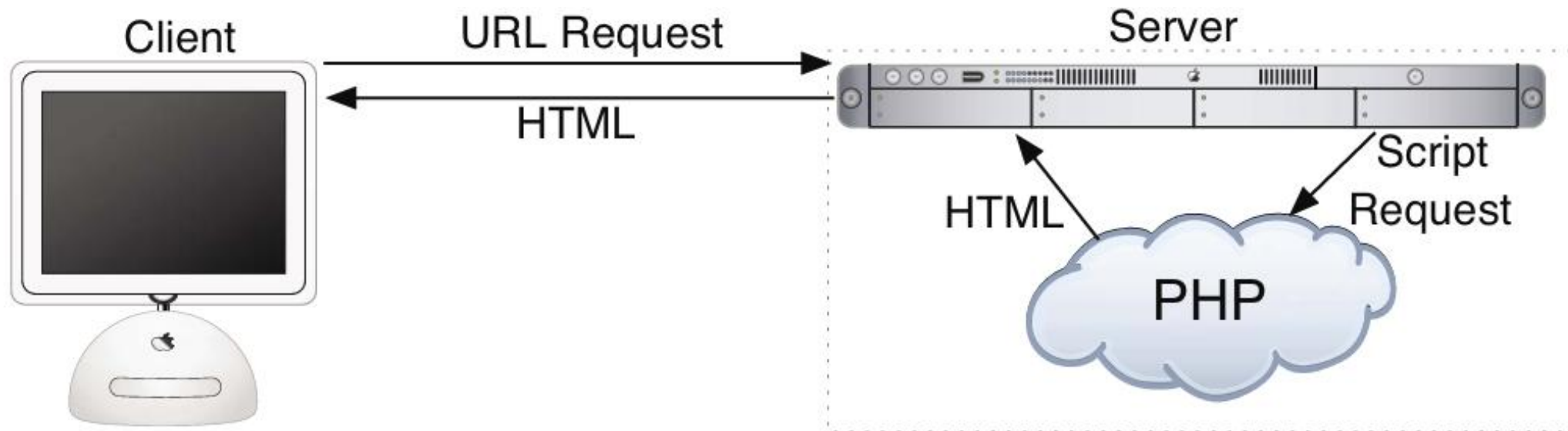
- PHP is a server-side, cross-platform technology, both descriptions being important.
 - Server side refers to the fact that PHP does occur on the server.
 - A web server application, like Apache or Microsoft's IIS (Internet Information Services), is required and all PHP scripts must be accessed
- PHP == 'Hypertext Preprocessor'
- Open-source, server-side scripting language
- Used to generate dynamic web-pages
- PHP scripts reside between reserved PHP tags
 - This allows the programmer to embed PHP scripts within HTML pages

Why use PHP?

- PHP is better, faster, and easier to learn than the alternatives.
- PHP is excellent performance, a tight integration with nearly every database available, stability, portability, and a nearly limitless feature set due to its extensibility.
- All of this comes at no cost (PHP is open source) and with a very manageable learning curve.

How PHP works

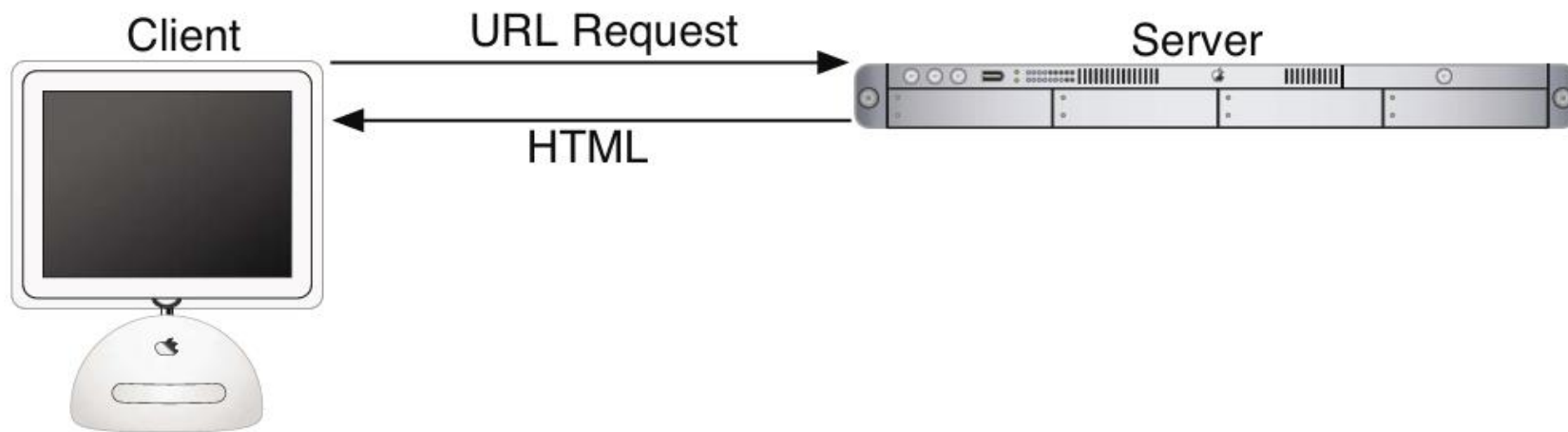
- The server sends web pages to the requesting visitors (you, the client, with your browser).
- When a visitor goes to a site written in PHP, the server reads the PHP code and then processes it according to its scripted directions.



D How PHP fits into the client/server model when a user requests a page.

How PHP works

- A static HTML site where, when a request is made, the server merely sends the HTML data to the browser and there is no server-side interpretation occurring **E**.
- Because no server-side action is required, you can run HTML pages in your browser without using a server at all



E The client/server process when a request for a static HTML page is made.

Basic Syntax

- A basic HTML5 page.

```
1  <!doctype html>
2  <html lang="en">
3  <head>
4      <meta charset="utf-8">
5      <title>Page Title</title>
6  </head>
7  <body>
8      <!-- Script 1.1 - template.html -->
9  </body>
10 </html>
```

To add PHP code to a page, place it within PHP tags:

```
<?php
?>
```

```
1  <!doctype html>
2  <html lang="en">
3  <head>
4      <meta charset="utf-8">
5      <title>Basic PHP Page</title>
6  </head>
7  <body>
8      <!-- Script 1.2 - first.php -->
9      <p>This is standard HTML.</p>
10 <?php
11 ?>
12 </body>
13 </html>
```

To make a basic PHP script

1. Create a new document in your text editor or IDE, to be named first.php
2. Create a basic HTML document
3. Before the closing body tag, insert the PHP tags
4. Save the file as first.php
5. Place the file in the proper directory of your web server.
6. Run first.php in your browser

Comments in PHP

- In HTML you can add comments using special tags:

`<!-- Comment goes here. -->`

- PHP supports three comment syntaxes.

`# This is a comment.`

`// This is also a comment.`

- Both of these cause PHP to ignore every thing that follows until the end of the line
- A third style allows comments to run over multiple lines:

`/* This is a longer comment
that spans two lines. */`

Sending Data to the Browser

- PHP has a number of built-in functions

```
echo 'Hello, world!';
```

```
echo "What's new?";
```

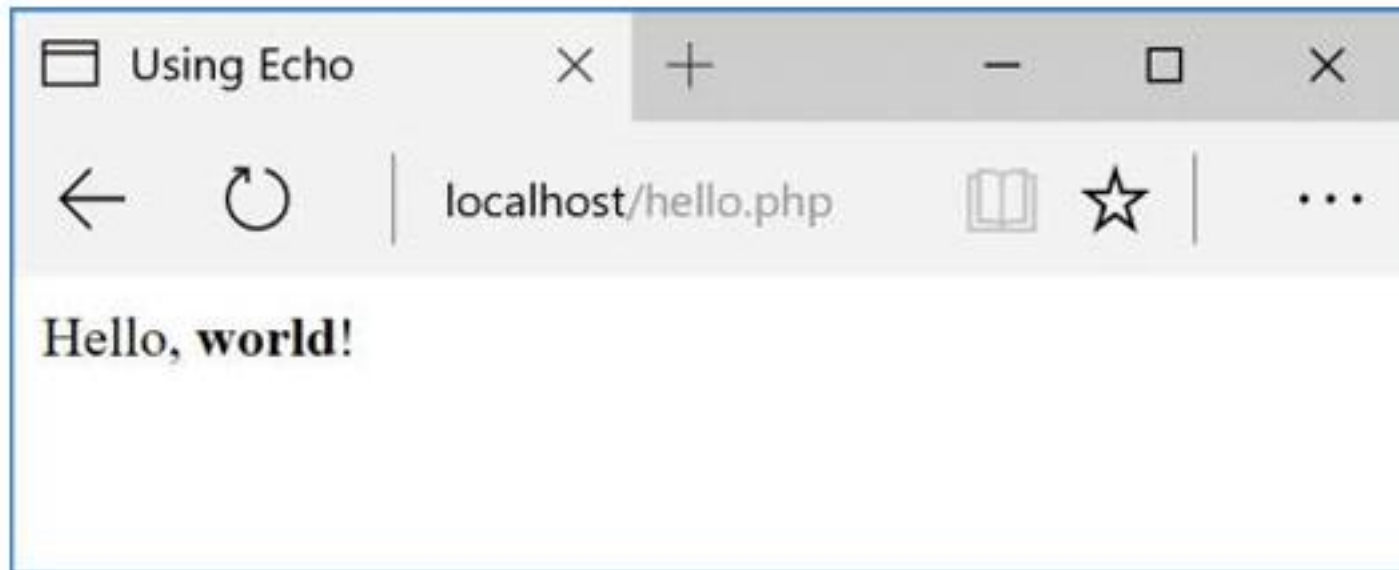
- You could use print instead if you prefer (the name more obviously indicates what it does):

```
print 'Hello, world!';
```

```
print "What's new?";
```

Exercise 01

- Print ' Welcome to PHP'



```
echo '<p>Hello, <strong>world</strong>!</p>';
```

Variables in PHP

- PHP variables must begin with a “\$” sign
- Case-sensitive (\$Foo != \$foo != \$fOo)
- Global and locally-scoped variables
- Global variables can be used anywhere
- Local variables restricted to a function or class
- Certain variable names reserved by PHP
- Form variables (\$_POST, \$_GET)
- Server variables (\$_SERVER)

Variable types

- **Integers:** are whole numbers, without a decimal point, like 4195.
- **Doubles:** are floating-point numbers, like 3.14159 or 49.1.
- **Booleans:** have only two possible values either true or false.
- **NULL:** is a special type that only has one value: NULL.
- **Strings:** are sequences of characters, like 'PHP supports string operations.'
- **Arrays:** are named and indexed collections of other values.
- **Objects:** are instances of programmer-defined classes, which can package up both other kinds of values and functions that are specific to the class.
- **Resources:** are special variables that hold references to resources external to PHP (such as database connections).

Variable usage

```
<?php
$foo = 25;           // Numerical variable
$bar = "Hello";      // String variable

$foo = ($foo * 7);    // Multiplies foo by 7
$bar = ($bar * 7);    // Invalid expression
?>
```


Echo

```
<?php
$foo = 25;           // Numerical variable
$bar = "Hello";      // String variable

echo $bar;           // Outputs Hello
echo $foo,$bar;       // Outputs 25Hello
echo "5x5=", $foo;    // Outputs 5x5=25
echo "5x5=$foo";      // Outputs 5x5=25
echo '5x5=$foo';      // Outputs 5x5=$foo
?>
```

- Notice how echo '5x5=\$foo' outputs \$foo rather than replacing it with 25
- Strings in single quotes (' ') are not interpreted or evaluated by PHP
- This is true for both variables and character escape-sequences (such as "\n" or "\\")

Arithmetic Operations

```
<?php
    $a=15;
    $b=30;
    $total=$a+$b;
    Print $total;
    Print "<p><h1>$total</h1>";
    // total is 45
?>
```

- $\$a - \b // subtraction
- $\$a * \b // multiplication
- $\$a / \b // division
- $\$a += 5$ // $\$a = \$a + 5$ Also works for $\ast =$ and $/ =$

Exercise 02

- Write a programme to add, subtract, multiply and divide two numbers

```
<head> <title>Arithmetical Operators</title> </head>
<body>
<?php
    $a = 42;
    $b = 20;
    $c = $a + $b;
    echo "Addition Operation Result: $c <br/>";
    $c = $a - $b;
    echo "Subtraction Operation Result: $c <br/>";
    $c = $a * $b;
    echo "Multiplication Operation Result: $c <br/>";
    $c = $a / $b;
    echo "Division Operation Result: $c <br/>";
    $c = $a % $b;
    echo "Modulus Operation Result: $c <br/>";
    $c = $a++;
    echo "Increment Operation Result: $c <br/>";
    $c = $a--;
    echo "Decrement Operation Result: $c <br/>";
?>
</body>
</html>
```

Thank you